

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)

Assessment Tool: Annexure A - Concept Mapping

Assessment Tool Weightage: 30%

Faculty: Dr. V. Parthipan

Code of Conduct:

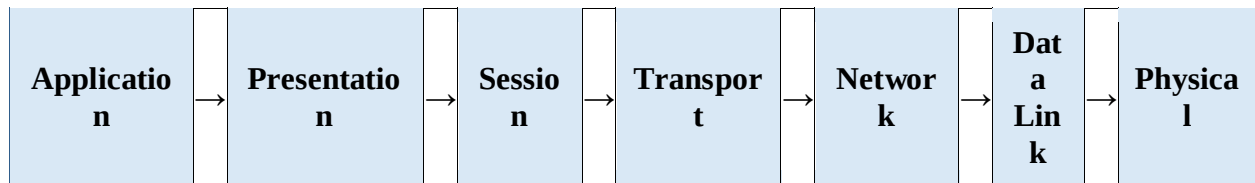
I **B Tharun Kumar / 192424356** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **B Tharun Kumar**

Question 1: Missing OSI Layers in the Concept Map

Given concept map:



The two missing layers are the Session Layer (between Presentation and Transport) and the Data Link Layer (between Network and Physical).

Analysis and Explanation:

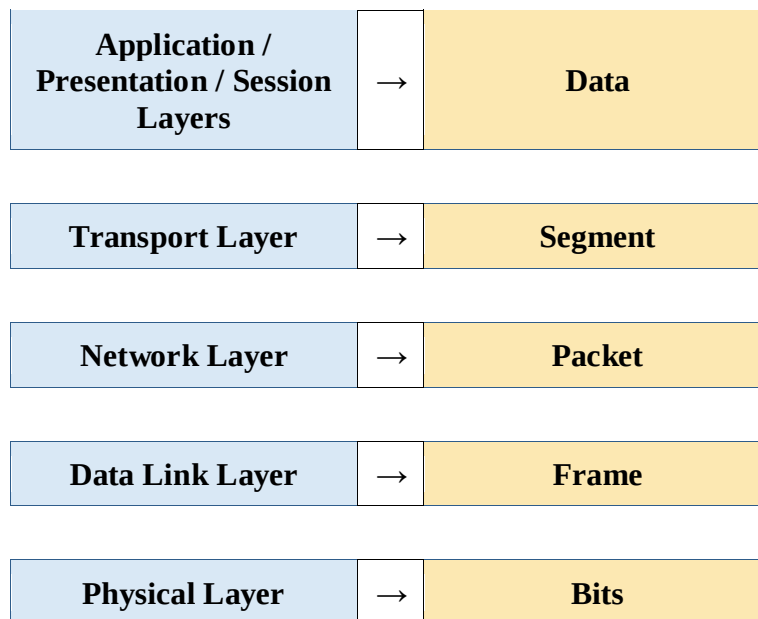
Session Layer: This layer is responsible for establishing, managing, and terminating sessions (dialogues) between two communicating applications. It handles synchronization by inserting checkpoints into the data stream, so that if a connection fails, communication can resume from the last checkpoint instead of restarting from scratch. It also manages dialogue control, deciding whether communication is half-duplex or full-duplex. Without the Session layer, applications would have no structured way to open, coordinate, and close long-running conversations reliably.

Data Link Layer: This layer takes packets received from the Network layer and frames them for transmission over the physical medium. It performs physical addressing using MAC addresses, ensuring that data reaches the correct device on a local network segment. It also carries out error detection (using techniques such as CRC) and flow control between two directly connected nodes, and manages media access control so that multiple devices sharing the same medium do not collide.

Together, these two layers are essential for reliable communication: the Session layer ensures that conversations between end applications are properly coordinated and recoverable, while the Data Link layer ensures that each individual hop of the transmission is correctly addressed, framed, and checked for errors before being placed on the physical medium. Removing either layer would break either the logical coordination of the conversation or the physical reliability of each transmitted frame.

Question 2: Concept Map of Data Units and OSI Layers

Concept Map:



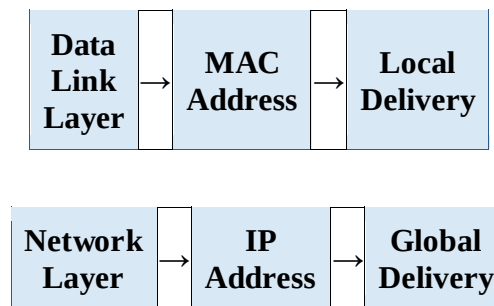
Analysis:

As data moves down the OSI stack from the sender, each layer adds its own header (and, at the Data Link layer, a trailer) to the data received from the layer above. This process is called encapsulation. Raw application data is first treated simply as Data; the Transport layer wraps it into a Segment by adding source/destination port numbers and sequencing information for reliability; the Network layer wraps the segment into a Packet by adding logical IP addressing for routing; the Data Link layer wraps the packet into a Frame by adding MAC addressing and error-checking information for delivery on the local link; and finally the Physical layer converts the frame into a raw stream of Bits (electrical, optical, or radio signals) for transmission over the medium.

At the receiving end, the reverse process, called decapsulation, occurs: bits are reassembled into a frame, the frame's header/trailer are stripped to recover the packet, the packet's header is stripped to recover the segment, and the segment's header is stripped to recover the original data, which is then passed up to the receiving application. This transformation of data units at each layer ensures that addressing, routing, error control, and delivery responsibilities are cleanly separated, which makes reliable end-to-end transmission possible even across complex, multi-hop networks.

Question 3: MAC Address vs IP Address — Local and Global Delivery

Concept Map:



Explanation:

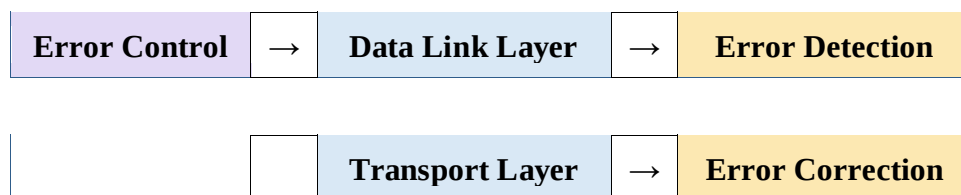
The Data Link layer uses the MAC (Media Access Control) address, a unique 48-bit physical address burned into a device's network interface card, to achieve local delivery. When a frame needs to travel from one device to another on the same local network segment (for example, from a computer to the nearest router, or between two devices on the same switch), the Data Link layer uses the MAC address to identify exactly which physical device on that segment should receive the frame. MAC addressing operates only within a single broadcast domain and does not extend across different networks.

The Network layer, in contrast, uses the IP (Internet Protocol) address to achieve global delivery. IP addresses are logical and hierarchical, allowing routers to determine the best path for a packet across multiple interconnected networks, potentially spanning the entire internet. The IP address identifies the ultimate source and destination hosts regardless of which physical network they are attached to.

Together, these two addressing schemes ensure complete, end-to-end data delivery: at every hop along the path, the Network layer's IP address tells the packet where it ultimately needs to go (global delivery), while the Data Link layer's MAC address tells each frame which physical device on the current segment should receive it next (local, hop-by-hop delivery). Routers repeatedly strip and rebuild the MAC-layer framing at each hop while preserving the IP-layer addressing, effectively relaying the packet from source to destination one local delivery at a time until the final global destination is reached.

Question 4: Error Control Across Data Link and Transport Layers

Concept Map:



Analysis:

The Data Link layer performs error detection at the level of individual frames on a single physical link. It typically uses techniques such as the Cyclic Redundancy Check (CRC) or a checksum, appending a small trailer to each frame. When a frame arrives at the receiving node, the receiver recalculates the check value and compares it with the one received; if they do not match, the frame is known to be corrupted and is usually simply discarded (or, in some link-layer protocols, retransmission at that hop is requested). This detection happens quickly and locally, close to the physical medium where transmission errors such as noise or interference are most likely to occur.

The Transport layer, on the other hand, is responsible for end-to-end error correction and reliability across the entire path from source to destination, which may involve many intermediate hops. Protocols such as TCP use sequence numbers, acknowledgements, checksums, and retransmission timers to detect lost, duplicated, or out-of-order segments and to request retransmission of only the missing or corrupted data, ultimately delivering a complete and correctly ordered byte stream to the receiving application.

The interaction between these two layers significantly improves overall reliability: the Data Link layer filters out most transmission errors close to their source, hop by hop, reducing the burden on higher layers, while the Transport layer provides a safety net that guarantees end-to-end correctness even if a corrupted frame slips through undetected at some hop, or if a packet is lost due to congestion or routing failure somewhere in the network. This layered, defense-in-depth approach to error handling is a key reason why data transmitted across large, unreliable networks like the internet still arrives intact at its destination.

Question 5: Concept Map for Web Browsing Using OSI Layers

Concept Map (Top-down flow when a user opens a webpage in a browser):

Application Layer	→	Browser generates an HTTP/HTTPS request for the webpage
Presentation Layer	→	Encrypts data using TLS/SSL and encodes/formats content
Session Layer	→	Establishes and maintains the session with the web server
Transport Layer	→	TCP breaks data into segments; ensures reliable, ordered delivery
Network Layer	→	IP addresses source and destination; routes packets across the internet
Data Link Layer	→	Frames the packet with MAC addressing for local hop delivery
Physical Layer	→	Transmits bits as electrical, optical, or wireless signals

Analysis:

When a user types a URL and presses Enter, the browser (Application layer) formulates an HTTP or HTTPS request. The Presentation layer ensures the data is properly encoded and, for HTTPS, encrypted via TLS so that it can be safely and correctly interpreted by the destination server. The Session layer keeps the connection to the web server organized, especially important for websites that require multiple exchanges or maintain login state. The Transport layer (typically TCP) breaks the request into segments, adds port numbers (such as port 443 for HTTPS), and guarantees reliable, in-order delivery through acknowledgements and retransmissions. The Network layer attaches IP addressing and determines the routing path across possibly dozens of intermediate routers to reach the web server. The Data Link layer frames the packet for each individual hop

using MAC addressing, and the Physical layer converts these frames into actual electrical, optical, or radio signals that travel across cables, fibre, or wireless links.

Because every layer depends on the correct functioning of the one below it, a failure at any single layer disrupts the entire process: if the Physical layer fails (e.g., a cut cable or no wireless signal), no bits can be transmitted at all, and every layer above it becomes irrelevant. If the Data Link layer fails (e.g., a MAC address conflict or a faulty switch port), frames cannot reach the correct local device even though the physical medium is intact. If the Network layer fails (e.g., a misconfigured router or missing route), packets cannot find a path to the destination even though the local link works. If the Transport layer fails (e.g., a blocked port or a broken TCP handshake), the segments cannot be reliably delivered even though packets are reaching the network. Finally, if the Application layer fails (e.g., the web server itself is down or the browser has a bug), the user sees an error even though every layer below successfully delivered the data. This demonstrates that reliable web browsing depends on the correct, coordinated functioning of all seven OSI layers, not any single layer in isolation.

Question 6: Troubleshooting Using the OSI Concept Map

Given concept map (status of each layer during the fault):

Physical Layer	Working (✓)
Data Link Layer	Working (✓)
Network Layer	Not Working (X)
Transport Layer	Not Working (X)
Application Layer	Not Working (X)

Analysis and Diagnosis:

The concept map shows that the Physical and Data Link layers are functioning correctly, while the Network layer and every layer above it are failing. Since the OSI layers are strictly hierarchical, with each layer depending on the successful operation of the layer(s) beneath it, the fact that the two lowest layers work correctly means that the cabling, network interface card, signal transmission, and local (same-segment) addressing and framing are all fine — the device can physically connect to the network and communicate with directly attached neighbours (such as a switch).

The failure begins at the Network layer, which strongly suggests the root cause lies in IP-level configuration or routing rather than in the physical connection or the local link. Typical causes at this point include: an incorrect or missing IP address/subnet mask, a missing or incorrect default gateway configuration, a failed or misconfigured router, or a routing loop/blackhole somewhere upstream. Because the Network layer cannot successfully deliver packets, the Transport layer has nothing to build a reliable connection on top of, so it also fails; and because the Transport layer cannot deliver data to the destination application, the Application layer likewise fails, even though the application software itself may be completely healthy.

This illustrates the core value of a layered concept map for troubleshooting: instead of guessing randomly across the whole system, a technician can test each layer from the bottom up (or use tools such as ping to test Network-layer reachability, given that Physical/Data Link are confirmed healthy) and immediately narrow the search to the Network layer, since that is the lowest layer at which failure first appears. The concept map turns an otherwise vague symptom ("the internet isn't working") into a precise, layer-by-layer diagnostic path, saving time and pointing directly to IP addressing or routing configuration as the most probable fix.